ALPACA AI TRADING AGENTS HACKATHON

AFTERIMAGE

GUAVA MANGO

Evidence-bound autonomous trading
for Alpaca paper accounts

+48.05R

2022–2026 PSEUDO-OOS RESEARCH

95 TRADES / PF 2.253

Strategy display name GuavaMango
Internal strategy ID shadow_clock_shock_reversal_v1

JUDGE DECK / SEP 2026

+48.05R across five positive annual folds

2022–2026 chronological pseudo-OOS research. Backtest evidence, not broker P&L.

+48.05R

NET RESULT

95 modeled trades

2.253

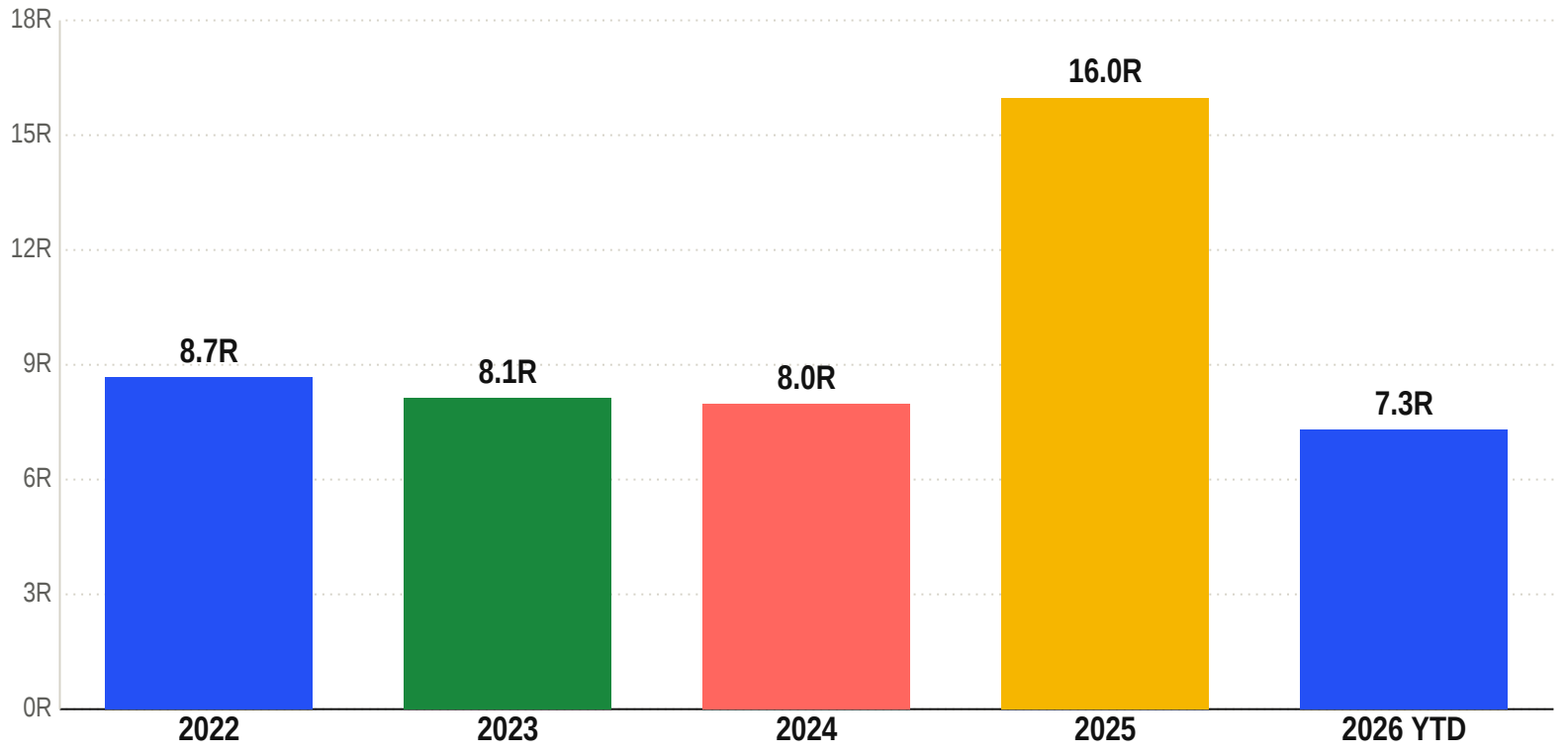
PROFIT FACTOR

+0.506R expectancy per trade

4.746R

MAX DRAWDOWN

72.34% positive active months



ANNUAL NET R

2026 through Sep 3

A stateful early-week ETF shock reversal

Long-only equity execution across a complete liquid ETF universe.

11 ETFs

QQQ	IWM	XLK	XLF
XLE	XLV	XLY	XLI
XLP	TLT	GLD	

Complete universe scoring prevents a partial-fetch winner from becoming a trade.



PREVIOUS CLOSE

Confirm established uptrend

TWO-DAY SHOCK

Pick strongest normalized decline

STATE CLOCK

Skipped Thu/Fri signals still advance state

NEXT OPEN

Rebase stop on executable quote

BY 10:00 ET

Submit equity bracket only if ≥3R

ENTRY CONTRACT

Next-session equity limit bracket · causal inputs · modeled target:stop ≥ 3.00

The result does not depend on one lucky setting

Every admission check passes; the failure modes remain visible.

TEST	RESULT	READ-THROUGH
Immediate parameter neighborhood	108 / 108 positive	Shock, close location, stop, and hold combinations
Modeled cost stress	+46.88R · PF 2.222	1.5× the base slippage and fee model
Annual folds	5 / 5 net positive	2022 through September 3, 2026
Underlying breadth	8 profitable ETFs	Edge is not concentrated in one ticker
Payoff floor	≥ 3.000013 target:stop	After modeled costs and cent rounding
Active-month breadth	72.34% positive	Across months with at least one modeled trade

ADMISSION RESULT

Strategy admitted. Research limits stay attached to the result.

The trade is the last step of the proof

A model can produce a persuasive explanation after it sees an outcome. AFTERIMAGE preserves the evidence chain that existed before capital authority.

One standard

A strategy may win, lose, or abstain. Every outcome remains visible and auditable.

01 **Freeze the claim**

Point-in-time market state and strategy version

02 **Challenge the evidence**

Information-separated agents can object or abstain

03 **Verify deterministic risk**

Costs, sizing, exposure, clock, payoff

04 **Sign exact authority**

Single-use warrant binds every order field

05 **Reconcile broker truth**

Alpaca state overrides local memory after ambiguity

EVERY DECISION SHIPS WITH A JUDGE-READABLE RECEIPT

The evidence survives the uncomfortable labels

A visible research window deserves an honest name.

WHAT THE TEST DID

01 **Authenticated Alpaca IEX history**

Cache: 2015–Sep 3, 2026; research partitions begin in 2018

02 **Causal fills and explicit costs**

Completed bars, next-open entries, adverse slippage, per-share fees

03 **Append-only attempt ledger**

Failed compression, flow, option, and reversal ideas remain visible

WHAT THE TEST CAN SAY

2022–2026 chronological pseudo-OOS

The period became visible during iterative research. It is not described as an untouched holdout.

Limits kept on the record

IEX is not SIP. Daily bars cannot reconstruct executable bid/ask quotes or market impact. Results are a backtest, not broker P&L.

ANTI-CHERRY-PICKING RECEIPT

Every candidate, parameter neighborhood, stress result, and hash stays inspectable.

One strategy socket, six controlled stages

The orchestrator stays unchanged when the strategy plugin changes.

01
MARKET DATA
Alpaca bars
point-in-time
snapshot

02
SIGNAL
Strategy plugin
frozen config

03
AI JURY
Strategist +
witnesses
strict evidence
packets

04
CRITIC
Prosecutor
challenge or
abstain

05
RISK ENGINE
3R gate + sizing
signed warrant

06
ALPACA
Official API +
CLI/MCP
broker truth

DETERMINISTIC AUTHORITY BOUNDARY

AI can recommend, challenge, or abstain. It cannot override data completeness, payoff math, account limits, or warrant validation.

Typed contracts / SQLite WAL / hash-chained audit / restart reconciliation

Different roles receive different evidence

The jury explains and challenges. It never owns execution authority.

01

STRATEGIST

`zai-org/GLM-5.3`

Builds the bounded case from the strategy packet

02

REGIME WITNESS

`zai-org/GLM-5.3-Flash`

Tests whether market context contradicts the setup

03

EXECUTION WITNESS

`zai-org/GLM-5.3-Flash`

Checks quote quality, liquidity, and implementation details

04

PROSECUTOR

`zai-org/GLM-5.3`

Looks for fatal evidence gaps and unsupported claims

`STRICT OUTPUT`	**Evidence citations, schema validation, redaction, deterministic sampling, timeout, and abstention.**
`CAPITAL AUTHORITY`	**Deterministic risk owns the final gate. Missing evidence or malformed output stops the trade.**

Equity execution with a modular Options Lab

The admitted GuavaMango strategy uses equity brackets. Options are a separate Level 3 platform capability.

DEPLOYED WINNER

Risk vector

Executable quote, fees, exit slippage, sizing, exposure, daily loss, drawdown

Signed warrant

Ed25519 binds account, evidence, intent, quantity, limit, risk vector, nonce, and expiry

Broker truth

Official Alpaca Trading API plus CLI/MCP. Idempotent IDs and restart reconciliation.

OPTIONS LAB

PLATFORM CAPABILITY

LEVEL 3

MULTI-LEG READY

Delta
Open interest
Spread filters
Option history

ACTIVE PATH

GuavaMango uses an equity limit bracket with target:stop ≥ 3R.

No active option order

PAPER INTEGRATION VERIFIED ALPACA PAPER · CONNECTED

Real account. Real market data. No forced trade.

An authenticated smoke run exercised the deployed path without manufacturing a setup.

01

ACCOUNT BOUND

ACTIVE · $100,000 equity · Level 3 options permission

02

DATA COMPLETE

2,783 current daily bars across all 11 ETFs

03

RULE HELD

Thursday returned NO_SIGNAL before order construction

READY FOR A NATURAL SETUP

Broker evidence begins only after a qualifying paper setup fills and exits.

Current exposure: 0 positions · 0 orders submitted

Four criteria.
One verifiable story.

AFTERIMAGE makes the strategy, refusal logic, risk authority, and broker state inspectable from one cockpit.

01 P&L

+48.05R research result with costs, drawdown, annual folds, and honest limits

02 TECHNICAL IMPLEMENTATION

Strategy socket, strict AI routes, signed authority, official Alpaca paths, restart reconciliation

03 CREATIVITY / ORIGINALITY

State clock, evidence-to-code binding, information-separated jury, visible Alpha Graveyard

04 PRESENTATION

A judge sees live account truth, every gate, options readiness, and the proof trail

Internal strategy ID

`shadow_clock_shock_reversal_v1`

PROOF BEFORE PERMISSION